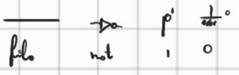
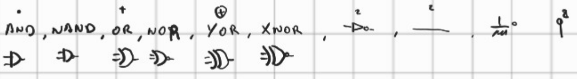


$y = f(x_1, x_2)$ — doppia variabile

$\overset{\cdot}{\text{AND}}$, NAND, $\overset{+}{\text{OR}}$, NOR, $\overset{\oplus}{\text{XOR}}$, XNOR, $\overset{2}{}$, $\overset{2}{}$, , q^2



x_1	x_2	$\cdot$
0	0	
0	1	
1	0	
1	1	

principio di enumerazione

$y = f(x_1, x_2, x_3)$ - 8 righe tab. verità (2^n pos)
- 2^8 funzioni (2^{2^n} funzioni)

Th Shannon (versione 1)

$y = f(x) =$

Th Shannon (versione 2)
il suo duale

$$y = f(x_1, x_2) = (\bar{x}_1 + \bar{x}_2 + f(1,1)) \cdot$$
$$(\bar{x}_1 + x_2 + f(1,0)) \cdot$$
$$(\bar{x}_1 + \bar{x}_2 + f(0,1)) \cdot$$
$$(x_1 + x_2 + f(0,0)) \cdot$$

A	B	C	y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

SH 1
conteno gli 1

$\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}C + AB\bar{C}$

$\bar{A}\bar{B}(\underbrace{\bar{C}+C}_{1}) + A\bar{B}C + AB\bar{C}$

$\bar{A}\bar{B} + A\bar{B}C + AB\bar{C}$

SH2
prendo gli 0

$\underbrace{(A+\bar{B}+C)\cdot(A+\bar{B}+\bar{C})}\cdot$
$\cdot(\bar{A}+B+C)\cdot(\bar{A}+\bar{B}+\bar{C})$

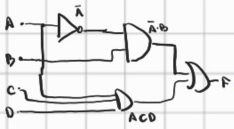
$\boxed{(A+\bar{B})} = \text{pippo}$

$(\text{pippo}+C)\cdot(\text{pippo}+\bar{C})$

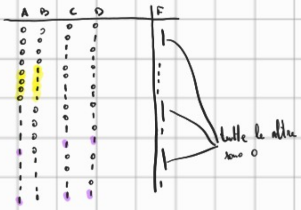
$\text{pippo} + \text{pippo}\cdot C + \text{pippo}\cdot\bar{C} + \cancel{0} = \text{pippo}+\text{pippo}$
$= \text{pippo}$

$(A+\bar{B}+C)\cdot(A+\bar{B}+\bar{C})\cdot(\bar{A}+B+C)\cdot(\bar{A}+\bar{B}+\bar{C})$

Esercizio



$F = \bar{A}B + \overline{ACD}$ — tutti = 1 — somma di prodotti
→ 4 variabili = 16 comb.


A
B
C
D
F
tutte le altre
sono 0

n° livelli della rete di partenza = 5

$F = ADC + \bar{C}(\bar{A}B + A\cdot(B+D))$ → $ADC + \bar{C}(\underbrace{\bar{A}B + AB}_{B(\underbrace{\bar{A}+A}_{1})} + AD)$ → $ADC + \bar{C}(B + AD)$

deve capire se può essere
SEMPLIFICATA
(stessa funz. ma + piccola)

$ADC + \bar{C}B + AD\bar{C}$

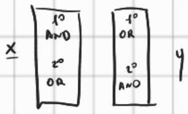
$AD(\underbrace{C+\bar{C}}_{1}) + \bar{C}B$

$\boxed{AD + \bar{C}B}$ n°. livelli = 2
numero massimo di porte logiche da incontrare

Th Shannon (1,2)

AND - OR

OR - AND

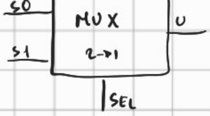


RETI A 2
LIVELLI
i NOT non contano nella rete

Multiplexer

Specifica testuale

U = S0 IF SEL=0



$U = S_1$ IF $SEL = 1$

Specifica tavola verità

SEL	S1	S0	
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Rappresentazione Matematica:

- SH1 (prendo 1)

$$\overline{SEL}\cdot\overline{S_1}\cdot S_0 + \overline{SEL}\cdot S_1\cdot S_0 + SEL\cdot S_1\cdot\overline{S_0} + SEL\cdot S_1\cdot S_0 =$$
$$= \overline{SEL}\, S_0 + SEL\cdot S_1$$

- SH2 (prendo 0)

$$(SEL + S_1 + S_0)\cdot(SEL + \overline{S_1} + S_0)\cdot(\overline{SEL} + S_1 + S_0)\cdot(\overline{SEL} + S_1 + \overline{S_0}) =$$
$$= (SEL + S_0)\cdot(\overline{SEL} + S_1)$$

Rappresentazione Circuitale:

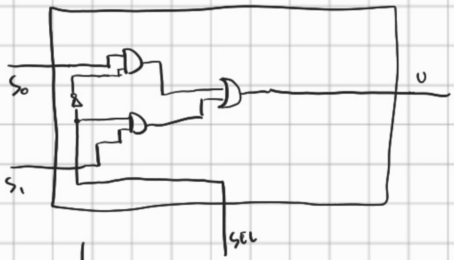


Diagramma Lineare

modo di esp. di una rete sequenziale

$F = A\cdot B$ (matematica)

(circuitale)

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

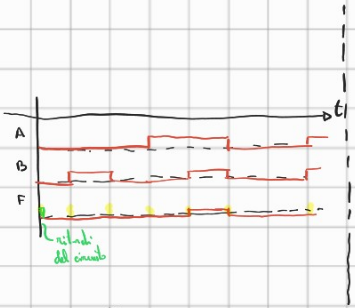
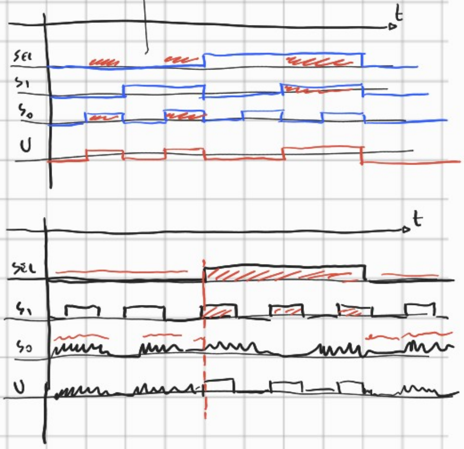


Diagramma Temporale

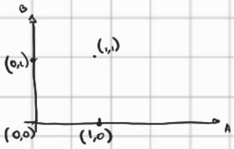
bisogna pensare che le cose siano OGGETTI che mi aiutano a fare qualcosa

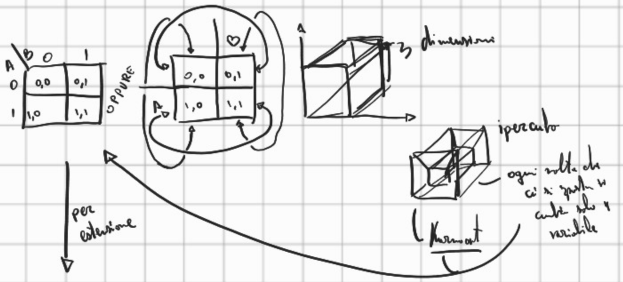


Tab. verità

→ Mappa di Karnaugh (rappres. grafica della tabella di verità)

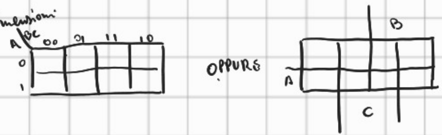
A	B	F
0	0	F(0,0)
0	1	F(0,1)
1	0	F(1,0)
1	1	F(1,1)



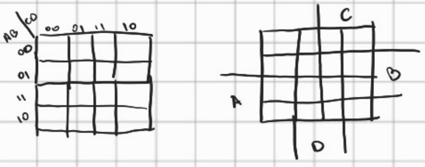


mappa a singola variabile

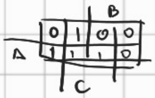
In 3 dimensioni



Mappa 4



A	B	C	U
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1



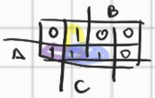
Ne faccio la sintesi

SHENNON 1

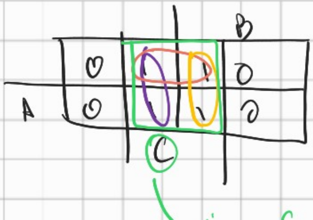
$$\overline{A}\,\overline{B}C + A\overline{B}\,\overline{C} + A\overline{B}C + ABC =$$
$$= \overline{A}\,\overline{B}C + A\overline{B} + ABC$$

rifaccio il tutto pensando che $A + A = A$

$$= \overline{A}\,\overline{B}C + \overline{A}\,\overline{B}C + A\overline{B}\,\overline{C} + A\overline{B}\,\overline{C} + A\overline{B}C + ABC$$
$$= \overline{B}C + A\overline{B} + AC$$



Mappa di Karnaugh



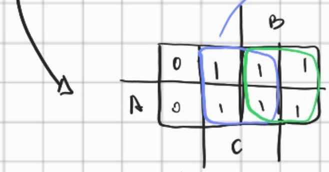
TH SA1

$$\overline{A}\,\overline{B}C + \overline{A}BC + A\overline{B}C + ABC$$

$$= \overline{B}C + BC$$

$$= C$$

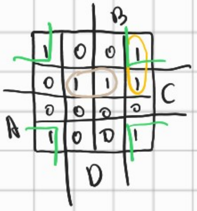
→ rimane C
perchè per assorbimento
vanno VIA



0	1	1	1
0	1	1	1

$U = C + B$

REGOLE:
1. coprire tutti gli uni
2. minimo n° di gruppi
3. gruppi + grandi possibile (2^n)



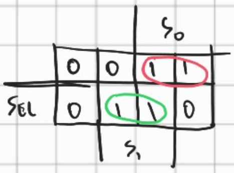
1	0	0	1
0	1	1	1
0	0	0	0
1	0	0	1

→ sintesi minima (non esiste solo 1° sintesi minima)

$U = \bar C \bar D + \bar A B \bar D + \bar A C D$

MUX 2→1

SEL	S_0	S_1	MUX
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



0	0	1	1
0	1	1	0

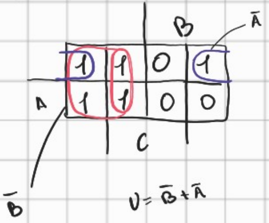
$$\boxed{U = SEL\, S_1 + \overline{SEL}\, S_0}$$

A	B	C	U
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

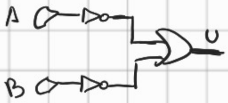
TR SH 1

$ABC + AB\bar C + A\bar B C + \bar A B C + \bar A B \bar C$

AB

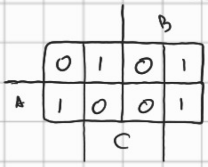


1	1	0	1
1	1	0	0



$U = \bar B + \bar A$

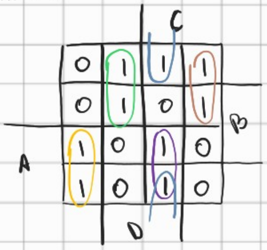
A	B	C	U
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0



0	1	0	1
1	0	0	1

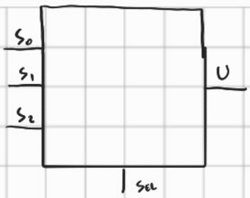
$\bar A \bar B C + \bar A B \bar C + A \bar B \bar C + A B \bar C = U$

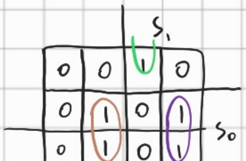
$\bar A + A \bar C = U$



0	1	1	1
0	1	0	1
1	0	1	0
1	0	1	0

$\bar B C D + A \bar C \bar D + \bar A \bar C D + ACD + \bar A C \bar D$

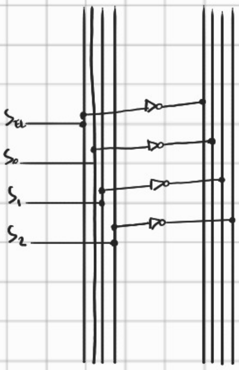




0	0	1	0
0	1	0	1
0	1	0	1

$S_0 S_1 \bar S_2 + S_{el} \bar S_0 \bar S_2 + \bar S_0 S_1 S_2 + S_0 \bar S_1 S_2$

S_EL
1
0
1
1
S_2


S_EL
S_0
S_1
S_2